

Question:

A 61-year-old woman is brought to an urgent care clinic by her husband with an altered mental status. The patient’s husband says that her symptoms onset acutely 24 hours ago. He says she suddenly started to feel excessively drowsy and quickly became altered. Past medical history is significant for renal transplant 18 months ago, well-managed with immunosuppressive medication. The vital signs include: temperature 39.4°C (103.0°F), blood pressure 85/50 mm Hg, pulse 135/min and respirations 24/min. On physical examination, the patient is arousable but non-responsive to commands. Oriented x 0. Glasgow Coma Scale (GCS) score is 10/15. Multiple black skin lesions are present on the trunk. Lesions are approximately 4 cm is diameter and round with a necrotic center forming an eschar.

Laboratory findings are significant for the following:

Hemoglobin 14.2 g/dL

WBC count 3,700/μL

Neutrophils 22%

Lymphocytes 52%

Monocytes 17%

Eosinophils 5%

Basophils 4%

Platelets 179,000/μL

BUN 15 mg/dL

Creatinine 0.8 mg/dL

Blood cultures are positive for Pseudomonas aeruginosa. Which of the following factors is most likely responsible for this patient’s condition?

Answer Choices:

- A. Failure of immunosuppressive therapy
- B. Decreased interleukin-2 levels
- C. DiGeorge Syndrome
- D. Decreased phagocytic cell count

Reasoning + Answer:

<think> Reasoning process here: The patient presents with symptoms of sepsis , as evidenced by fever , tachycardia , and altered mental status . The presence of multiple black skin lesions with necrotic centers is highly suggestive of cutaneous necrotizing fasc i itis , a condition often associated with bacterial infections . The laboratory findings of leuk open ia , specifically a low neutrophil count , further support this diagnosis . Additionally , the positive blood cultures for Pseud omon as aerugin osa , a bacter ium commonly implicated in cutaneous nec rot izing fasc i itis , provide a clear indication of the underlying cause . The patient's immun osup pressed status due to renal transplant and immun osup pressive medication could potentially increase the risk of infection , but it is not the most likely factor in this scenario . Decre ased interleuk in - 2 levels and DiGeorge Syndrome are not directly related to the patient's condition . </think>< answer> A . Failure of immun osup pressive therapy </ answer>